

Anthony J. Barrows

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EDUCATION

The University of Vermont

M.S. in Complex Systems and Data Science

Burlington, VT

Aug. 2019– May 2021

University of Hartford

B.A. in Psychology; Minor in Criminal Justice

Hartford, CT

Aug. 2013 – May 2016

EXPERIENCE

Vermont Center on Behavior and Health

Burlington, VT

Research Data Analyst

June 2021 – Present

- Independently developed automated clinical trial dashboards that eliminated time-consuming manual reports.
- Engineered data cleaning and transformation pipelines making trial data available for immediate use, reducing analysis delays.
- Produce analytical reports using open-source code making results transparent, reproducible, and accessible.
- Design, program, and maintain secure databases for storing sensitive information.
- Identify inefficiencies and implement solutions using automation to improve data collection methods.
- Collaborate with statisticians to manage and analyze data for publication, focusing on quality tables and figures.
- Review study data to ensure compliance with research protocols, training staff on best practices.
- Coordinate with outside vendors to oversee the design and maintenance of digital data collection systems.
- Prepare time-sensitive reports for regulatory boards and funding agencies including IRBs, DSMBs, and Federal collaborators.

Lead Research Assistant

October 2018 – June 2021

- Oversaw subject recruitment team, and implemented procedures to improve response and retention.
- Coordinated with study principal investigators to assess clinical challenges and quickly find effective solutions.
- Trained research staff to conduct participant sessions and use specialized equipment.

Research Assistant

October 2016 – October 2018

- Implemented NIDA/FDA grant-funded study on long term exposure to low nicotine cigarettes.
- Independently coordinated physiological and psychological testing (e.g., breath and urine analysis, psychiatric inventory), distributing study product, compensating participants, and ensuring adherence to IRB, DSMB and NIH-NIDA guidelines.

PROJECTS

Enrollment Report | *R, R Shiny, REDCap API, Linux CLI*

January 2020 – Present

- Create and maintain an interactive data dashboard for instantaneous and longitudinal clinical trial progress reporting.
- Interface with multiple APIs and fundamentally different data sources to retrieve information.
- Employ principles of tidy data to efficiently wrangle longitudinal sources for instantaneous aggregation.
- Build filtering tools to allow for flexible, live data exploration, prompting new insight from advanced visualizations and summaries.

TECHNICAL SKILLS

Languages: R, Python, L^AT_EX, Markdown

Frameworks: R Shiny, R Markdown, Microsoft Access, Linux CLI, REDCap API

Developer Tools: R Studio, Jupyter, VS Code

Version Control: Git, GitHub

Libraries: tidyverse, ggplot2, randomLCA, Plotly, pandas, NumPy, Matplotlib, scikit-learn

Software: Microsoft Excel, REDCap (Project Design), Final Cut Pro

AFFILIATIONS

BDI Enterprises, LLC

East Montpelier, VT

Founder, Limited Partner

Oct. 2020 – Present

- Organize educational jazz summer camps for adult musicians.
- Maintain business website, produce promotional videos and recordings.